

S. aureus Bacteremia

The five core interventions — for IM residents

Step 1 — The five core interventions

S. aureus in blood is NEVER a contaminant

#	Core intervention
1	ID consultation — independently lowers mortality
2	Echocardiography to evaluate for endocarditis
3	Repeat blood cultures q48–72h to document clearance
4	Source control — remove lines, drain foci
5	Appropriate drug & duration (complicated vs uncomplicated)

PEARL Call ID on every SAB — the consult itself is a mortality intervention.

Step 2 — Right drug: MSSA vs MRSA

Empiric vancomycin, then de-escalate

Scenario	Therapy
Empiric (pending)	Vancomycin (covers MRSA)
MSSA	Cefazolin OR antistaph penicillin (nafcillin/oxacillin)
MSSA on vancomycin	De-escalate — vancomycin is INFERIOR for MSSA
MRSA	Vancomycin (AUC-guided) or daptomycin

PEARL The classic error: leaving a confirmed MSSA on vancomycin for the whole course.

Step 3 — Complicated vs uncomplicated → duration

Uncomplicated needs ALL four criteria

Uncomplicated needs ALL	Complicated if ANY
Negative follow-up cultures	Persistent positive cultures
Defervescence within 72 h	Endocarditis
No endocarditis / prosthesis	Implanted hardware/prosthesis
No metastatic foci	Metastatic focus (bone, joint, abscess)

PEARL Uncomplicated ≥14 d IV from first negative culture; complicated 4–6 wk+. Count from clearance.

Hunt for metastatic foci

- Endocarditis — new murmur, peripheral stigmata
- Vertebral osteomyelitis / discitis — back pain → MRI
- Epidural abscess — back pain + neuro signs
- Septic arthritis — joint pain/effusion
- Psoas / splenic abscess, septic emboli
- TEE > TTE — use for prosthesis/device, persistent bacteremia, community-onset, unclear source

PEARL New back pain + persistent bacteremia = spine focus until imaging proves otherwise.

What to avoid

DO

- All five core interventions, every case
- De-escalate MSSA to cefazolin/antistaph penicillin
- Document clearance with repeat cultures
- Time duration from first negative culture

AVOID

- Calling *S. aureus* a contaminant
- Treating it like a simple line/UTI
- Keeping confirmed MSSA on vancomycin
- Short course without uncomplicated criteria

PEARL No documented clearance → you can't start counting and can't call it uncomplicated.

Take-home points

- *S. aureus* in blood is never a contaminant — five core interventions every time
- ID consult, echo, repeat cultures, source control, drug/duration
- Empiric vancomycin → de-escalate MSSA to cefazolin/antistaph penicillin
- Uncomplicated (all 4 criteria) = 14 d from clearance; complicated = 4–6 wk+
- Hunt metastatic foci; TEE when higher-risk or bacteremia persists